Connie Xu

Contact Information

8 Story Street, Suite 380
Cambridge, MA 02139

connniexu@g.harvard.edu
Mobile: (240) 429-2307
Citizenship: USA

Education

Harvard University — 2021–2027 (expected)
Ph.D. in Health Policy and Economics

University of Chicago — 2015–2019
B.A. in Economics (with Honors); B.A. in Statistics

Teaching Experience

Harvard Medical School — 2025
Small Group Leader for Essentials of the Profession Course – Nursing Home Module

Harvard Kennedy School — 2024
Teaching Fellow for Professor Timothy Layton, Empirical Methods II (Econometrics)

Relevant Positions

Harvard Business School — 2022–2024
Graduate Research Assistant to Professor Amitabh Chandra

Harvard Medical School — 2022–2024
Graduate Research Assistant to Professor Timothy Layton

Massachusetts Institute of Technology — 2019–2021
Pre-doctoral Research Fellow to Professor Amy Finkelstein

Research Papers

Scientific Input Costs in University Life Sciences Research, *with Ruby Zhang*
[Job Market Paper]

The real value of a research grant depends not only on the dollars it provides but on the prices those dollars must pay for laboratory inputs. We study how principal investigators (PIs) absorb higher input prices when their grant budgets are fixed in dollars and their personnel commitments cannot be adjusted within the funding cycle. Most of a grant is committed in advance to multi-year personnel contracts, so consumables such as reagents and enzymes are the only margin a PI can adjust, even as the firms supplying them have consolidated. We assemble novel procurement records of million purchases across universities from to , capturing what each PI buys, in what quantities, and at what prices. We study the merger of Thermo Fisher Scientific and Life Technologies as an exogenous shock to input prices. A difference-in-differences design estimates price increases of approximately percent in affected product markets with little quantity response, indicating highly inelastic demand. We then trace these cost shocks through an author-year panel of U.S. life sciences PIs. The average PI faces a percent increase in consumables costs and publishes percent fewer papers, an implied elasticity of output with respect to consumables costs of approximately . The decline appears concentrated in less-cited, lower-tier work, while output in the highest-impact journals and citation-weighted output show no significant change. The real value of public research funding therefore depends on the prices science pays for its inputs, not only the dollars that fund it.

The Returns to Science of Centrally Provisioned GPUs and Optimal Allocation, *with Ruby Zhang*

The rapid growth of AI research and the high cost of GPU infrastructure raise important questions about how federal funding for computing should be allocated. This project examines the scientific productivity impacts of centrally provisioned GPU clusters at universities. We structurally estimate a dynamic production function to causally determine the elasticity of AI research output with respect to central GPU cluster capacity, using the proxy variable approach of Ackerberg et al. (2015) with cloud computing expenditure as our proxy. We leverage novel data sources, including manually collected panel data on cluster hardware for the top 100 universities and purchase order data from GovSpend. With production function estimates in hand, we model optimal NSF funding allocation as a tradeoff between efficiency and equity, allowing us to identify where marginal returns to infrastructure investment would be greatest.

Person and Place Effects in Scientific Discovery, *with Amitabh Chandra*

How much of a scientist's research output is attributable to individual talent, and how much to the institution where they work? Because markets systematically under-provide fundamental science, the answer matters for how societies allocate the resources that substitute for missing market incentives, how the scientific community evaluates researchers for promotion and grants, and whether the current distribution of scientists across institutions maximizes knowledge production. We study this question using citation-weighted publications in leading life-science journals and a movers design that exploits variation in the output of nearly 40,000 scientists who change institutions. We find that 50 to 60 percent of the variation in a scientist's research output is attributable to institutional factors. After correcting for limited mobility bias, we document positive assortative matching between scientists and institutions (a correlation of 0.3 between person and place effects), a necessary condition for maximizing aggregate knowledge production under the multiplicative structure of our model. Around two-thirds of the institutional effect is explained by the presence of star researchers. A scientist's output responds to both the total output of their institution and its funding level, but not to the size of the local research cluster, suggesting that institutional effects operate through the human capital environment within an institution rather than through agglomeration or research funding alone. The symmetry of effects for upward and downward movers implies that institutional advantages do not permanently transfer to scientists who leave. These effects have not diminished despite technologies facilitating remote collaboration, are robust to detailed field-by-year controls, and raise the possibility that similar patterns extend to other areas of fundamental science.

Research in Progress

The Ripple Effects of Health Care Entry Regulation, *with Yunan Ji and Parker Rogers*
We examine how regulatory entry costs for healthcare inputs shape market structure, pricing, and bargaining dynamics throughout the healthcare supply chain. Using a difference-in-differences approach exploiting the FDA's 2016 removal of pre-market clearance requirements for certain medical devices, we find increased manufacturer entry and lower negotiated device prices between manufacturers and hospitals. These reduced input prices propagate downstream, modestly decreasing procedure prices negotiated with insurers and ultimately lowering patient out-of-pocket costs. We then develop and will estimate a structural model that jointly captures manufacturer entry decisions and simultaneous bargaining between hospitals, manufacturers, and insurers. Our proposed counterfactual simulations will quantify how tightening entry regulations or increasing horizontal consolidation would ripple through the supply chain.

Invited Presentations

The European Association for Research in Industrial Economics Conference	August 2026
American Society of Health Economists Conference	June 2026
Wharton Innovation Doctoral Symposium	March 2026
NBER Assessing the U.S. Medical Innovation System	March 2026
NBER Place-Based Policies and Entrepreneurship	December 2024

Professional Activities

Referee: JAMA	
Organizer: Harvard Health Care Policy Health Economics Seminar	2023–2025
Peer Mentor: Harvard Health Policy PhD Program	2022–2025
Affiliations: Institute for Quantitative Social Sciences	2024–present

Fellowships, Honors, and Awards

NBER Fiscal and Economic Effects of Innovation and Productivity Policies Fellow	2026–2027
National Science Foundation Graduate Research Fellowship Program	2024–2027
Harvard Kennedy School Distinction in Student Teaching Award	2024
NBER Pre-Doctoral Fellowship in Aging & Health	2023–2024
Agency for Healthcare Research and Quality T32 Trainee	2021–2023
Phi Beta Kappa	2019
UChicago Careers in Health Professions Health Policy Scholar	2016–2019

Research Grants

John and Kiendl Gordon Economics Support Fund, *with Ruby Zhang*	2025–2026
The Lab for Economic Applications and Policy Award, *with Ruby Zhang*	2024–2025
NBER Center for Aging and Health Research Pilot Grant, *with Amitabh Chandra and Timothy Layton*	2023–2024